

Nepal PreTST 2024 Solutions

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§1 Problems

Problem 1.1. Nirajan is trapped in a magical dungeon. He has infinitely many magical cards with arbitrary MPs (Mana Points) which is always an integer $\mathbb{Z}$. To escape, he must give the dungeon keeper some magical cards whose MPs add up to an integer with at least 2024 divisors. Can Nirajan always escape?

Problem 1.2. Let, $S = \sum_{i=1}^k n_i^2$. Prove that for $n_i \in \mathbb{R}^+$

$$\sum_{i=1}^k \frac{n_i}{S - n_i^2} \geq \frac{4}{n_1 + n_2 + \cdots + n_k}$$

Problem 1.3. Let ABC be an acute triangle and H be its orthocenter. Let E be the foot of the altitude from C to AB , F be the foot of the altitude from B to AC . Let $G \neq H$ be the intersection of the circles (AEF) and (BHC) . Prove that AG bisects BC .

Problem 1.4. Find all integer/s n such that $\frac{5^n - 1}{3}$ is a prime or a perfect square of an integer.

§2 Solutions

§2.1 Problem 1, proposed by Vlad Spătaru(Romania)

Problem 2.1. Nirajan is trapped in a magical dungeon. He has infinitely many magical cards with arbitrary MPs(Mana Points) which is always an integer $\mathbb{Z}$. To escape, he must give the dungeon keeper some magical cards whose MPs add up to an integer with at least 2024 divisors. Can Nirajan always escape?

Proof.

□

§2.2 Problem 2, proposed by Kang Taeyoung(South Korea)

Problem 2.2. Let, $S = \sum_{i=1}^k n_i^2$. Prove that for $n_i \in \mathbb{R}^+$

$$\sum_{i=1}^k \frac{n_i}{S - n_i^2} \geq \frac{4}{n_1 + n_2 + \cdots + n_k}$$

Proof. By Cauchy-Schwarz, we get:

$$\left(\sum_{i=1}^k \frac{n_i}{S - n_i^2} \right) \left(\sum_{i=1}^k n_i \right) \geq \left(\sum_{i=1}^k \sqrt{\frac{n_i^2}{S - n_i^2}} \right)^2$$

From AM-GM, we get:

$$\frac{S - n_i^2}{n_i^2} + 1 \geq 2\sqrt{\frac{S - n_i^2}{n_i^2}}$$

Which, upon flipping gives us

$$\sqrt{\frac{n_i^2}{S - n_i^2}} \geq \frac{2n_i^2}{S}$$

Therefore, summing from 1 to k gives,

$$\left(\sum_{i=1}^k \sqrt{\frac{n_i^2}{S - n_i^2}} \right) \geq \frac{4}{S^2} \left(\sum_{i=1}^k n_i^2 \right)^2 = 4$$

And we're done. □

§2.3 Problem 3, proposed by Kang Taeyoung(South Korea)

Problem 2.3. Let ABC be an acute triangle and H be its orthocenter. Let E be the foot of the altitude from C to AB , F be the foot of the altitude from B to AC . Let $G \neq H$ be the intersection of the circles (AEF) and (BHC) . Prove that AG bisects BC .

Proof.

□

§2.4 Problem 4, proposed by Prajit Adhikari(Nepal)

Problem 2.4. Find all integer/s n such that $\frac{5^n - 1}{3}$ is a prime or a perfect square of an integer.

Remark 2.5. The easiest problem on a Nepal PreTST, ever. Enjoy this solution I did over a year ago, when I barely knew number theory.

Proof. We claim the only answer is $n = 0$, which clearly works. We now prove no $n \geq 1$ exists (obviously $n < 0$ doesn't work). First, let p denote any prime, then,

$$\frac{5^n - 1}{3} = p$$

Taking modulo 4 gives us $p \equiv 0 \pmod{4}$, clearly ridiculous. Further let k denote any perfect square, so

$$5^n - 1 = 3k$$

Giving $-1 \equiv 3k \pmod{5}$ which again is absurd. Therefore, $n = 0$ is the only solution. $\square$